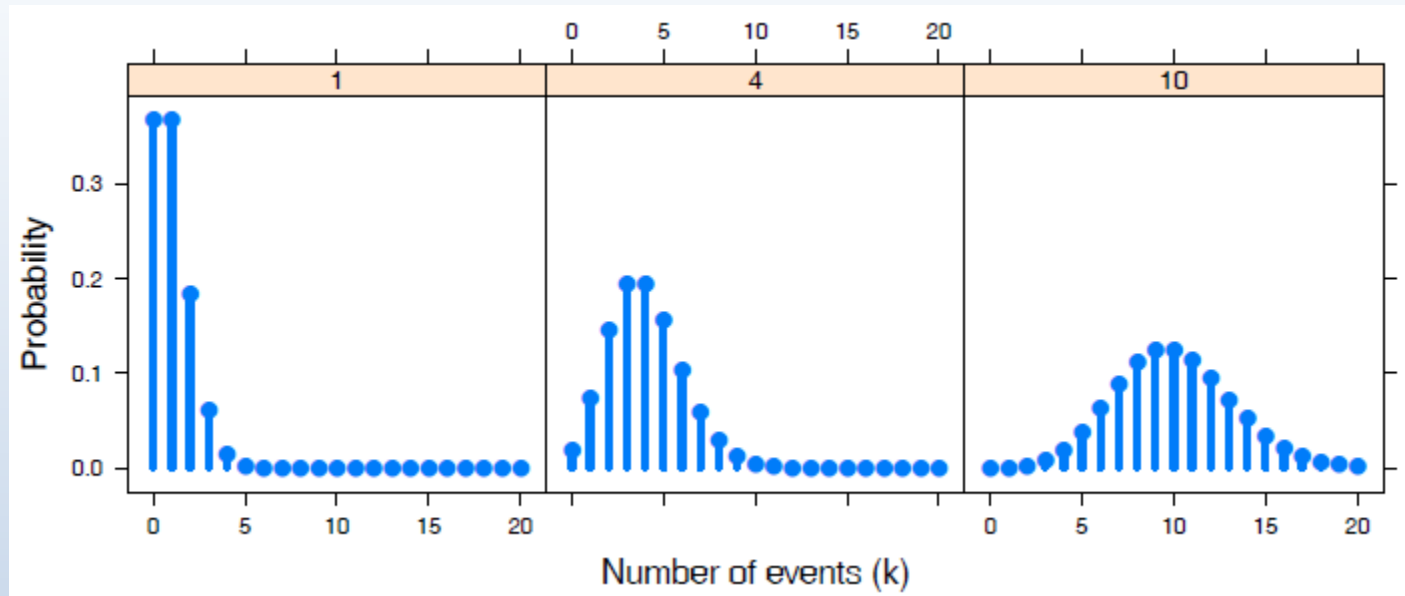
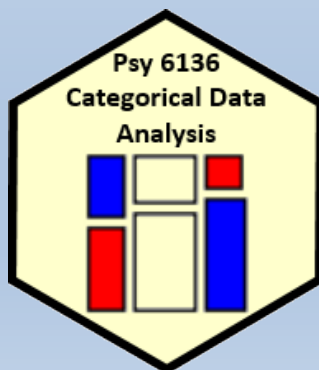




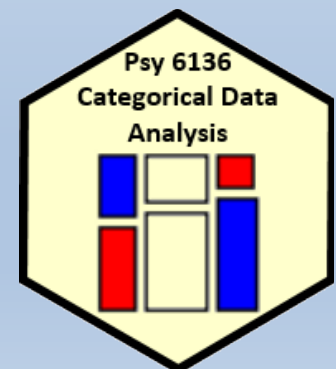
# Discrete distributions



Michael Friendly

Psych 6136

<http://friendly.github.io/psy6136>



# Discrete distributions: Basic ideas

- Quantitative data: often assumed Normal ( $\mu, \sigma^2$ ) – unreasonable for CDA
- Binomial, Poisson, Negative binomial, ... are the building blocks for CDA
- They form the basis for modeling techniques
  - logistic regression, generalized linear models, Poisson regression
- Data:
  - outcome variable ( $k = 0, 1, 2, \dots$ )
  - counts of occurrences ( $n_k$ ): accidents, words in text, males in families of size  $k$

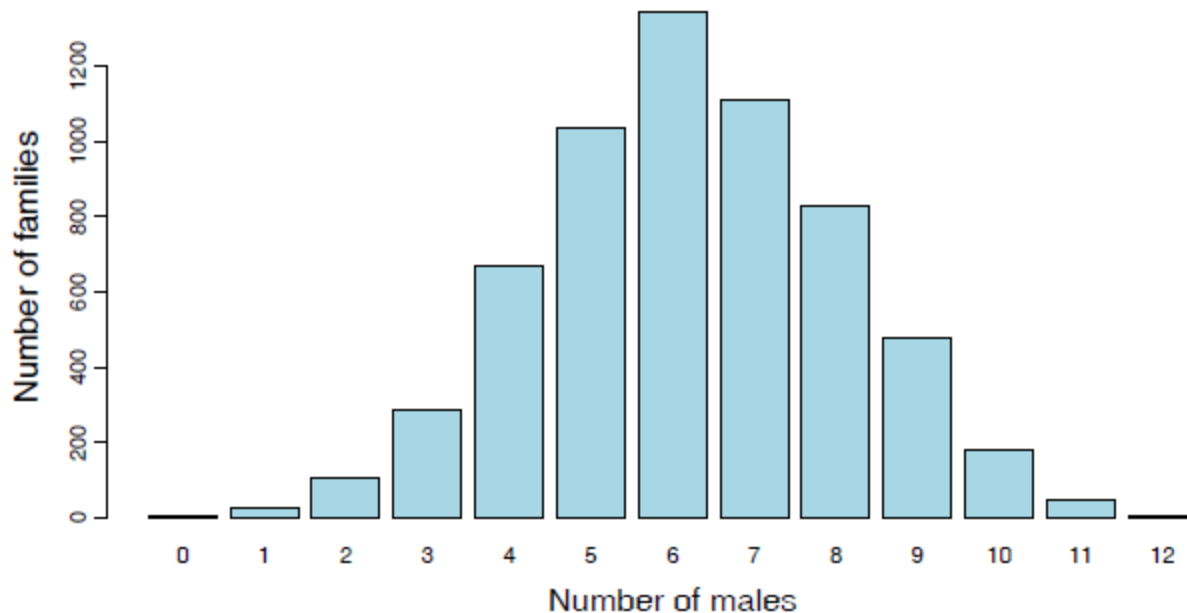
# Examples: binomial

Human sex ratio (Geissler, 1889): Is there evidence that  $\Pr(\text{male}) = 0.5$ ?

## Saxony families

Saxony families with 12 children having  $k = 0, 1, \dots, 12$  sons.

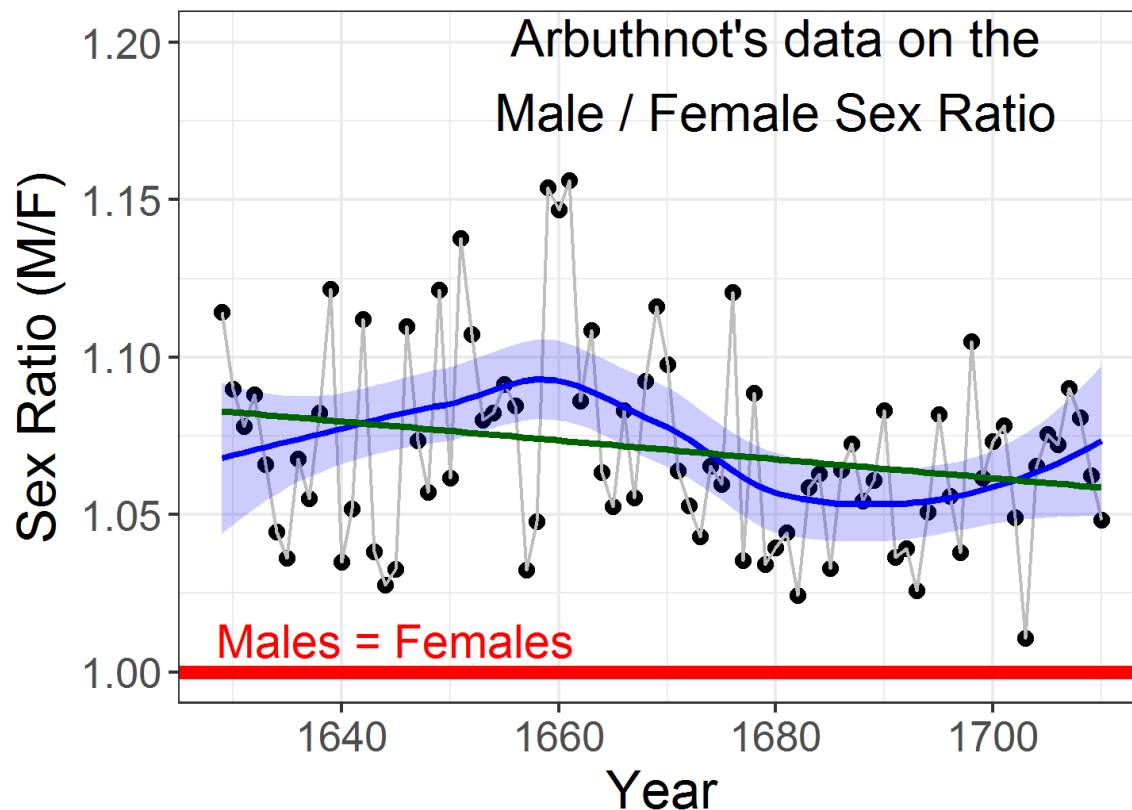
| $k$   | 0 | 1  | 2   | 3   | 4   | 5    | 6    | 7    | 8   | 9   | 10  | 11 | 12 |
|-------|---|----|-----|-----|-----|------|------|------|-----|-----|-----|----|----|
| $n_k$ | 3 | 24 | 104 | 286 | 670 | 1033 | 1343 | 1112 | 829 | 478 | 181 | 45 | 7  |



# Why did this matter? -- Arbuthot

John Arbuthot (1710) had gathered data on the ratios of male to female christenings in London from 1629-1710, 81 years

- He showed that in EVERY year Males > Females
- Calculated the FIRST significance test: like tossing 81 coins, and getting all heads
- Argued that the result must be due to divine providence: God likes more boys than girls!



Data: HistData::Arbuthnot

ggplot script:

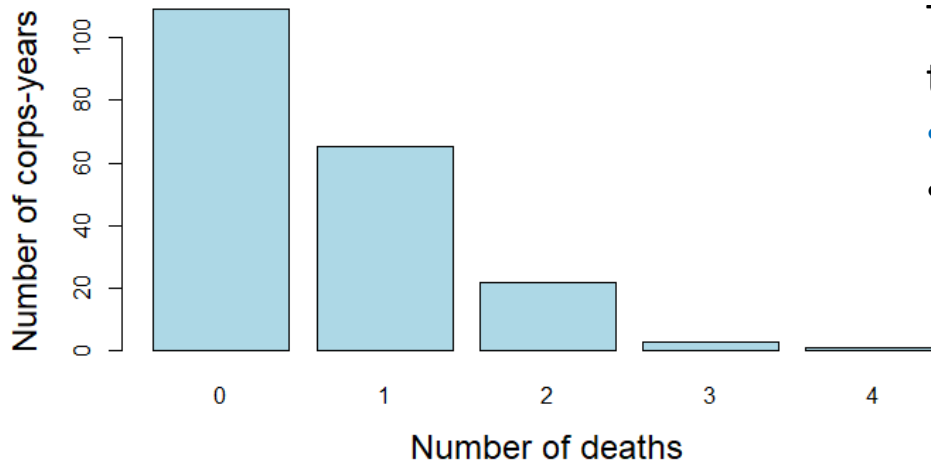
<https://friendly.github.io/6135/R/arbuthtnot-gg.html>

# Example: Poisson

L. Von Bortkiewicz (1898) tallied the numbers of deaths by horse or mule kicks in 10 corps of the Prussian army over 20 years, → 200 corps-years

- In how many corps-years were there 0, 1, 2, ... deaths?
- This is among the earliest examples of a Poisson distribution

```
> data(HorseKicks, package="vcd")  
> HorseKicks  
nDeaths  
  0    1    2    3    4  
109   65   22    3    1
```



The Poisson distribution arises as that of the probability of 0, 1, 2, ...

- Rare events, that
- Occur with constant probability

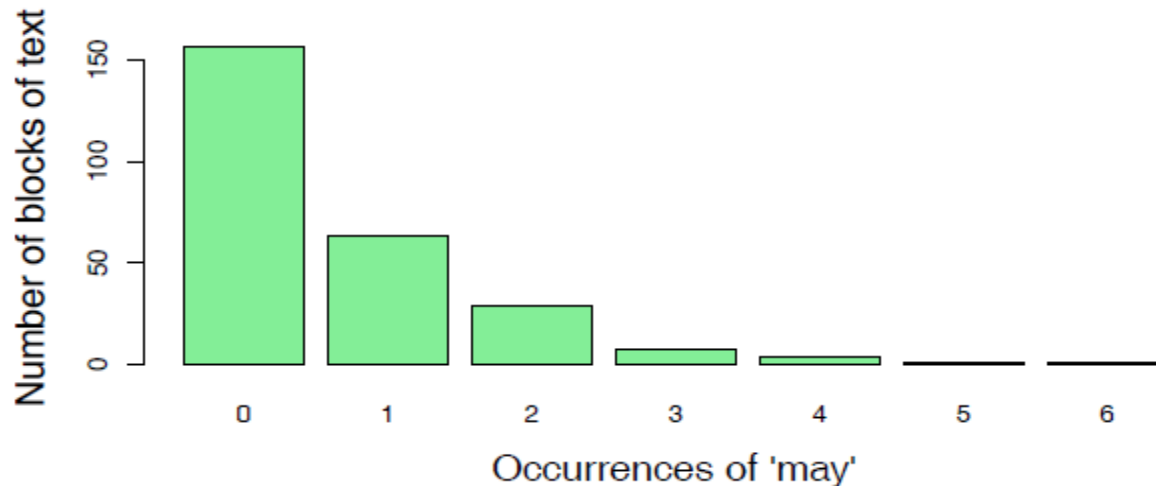
# Examples: count data

## Federalist papers: Disputed authorship

- 77 essays by Alexander Hamilton, John Jay, James Madison to persuade voters to ratify the US constitution, all signed with pseudonym “Publius”
  - Who wrote each?
  - 65 known, 12 disputed (H & M both claimed sole authorship)
- Mosteller & Wallace (1984): analysis of frequency dist<sup>n</sup>s of key “marker” words: from, **may**, whilst, ...
- e.g., blocks of 200 words: occurrences ( $k$ ) of “may” in how many blocks ( $n_k$ )

```
> data(Federalist, package = "vcd")
> Federalist
nMay
  0    1    2    3    4    5    6
156  63  29   8   4   1   1
```

# Count data: models



For each word (“from”, “may”, “whilst”, ...)

- Fit a probability model [ $\text{Poisson}(\lambda)$ ,  $\text{NegBin}(\lambda, p)$ ]
- Estimate parameters  $(\lambda, p)$
- → Calculate log Odds (Hamilton vs. Madison)
- → All 12 disputed papers most likely written by **Madison**

(pioneered the use of cross-validation to assess model fit)

# Example: Type-token distributions

- Basic count,  $k$ : number of “types”; frequency,  $n_k$ : number of instances observed
  - Frequencies of distinct words in a book or literary corpus
  - Number of subjects listing words as members of the semantic category “fruit”
  - Distinct species of animals caught in traps
- Differs from other distributions in that the frequency for  $k = 0$  is *unobserved*
- Distribution is often extremely skewed (J-shaped)

Table: Number of butterfly species  $n_k$  for which  $k$  individuals were collected

|                     |     |    |    |    |    |    |    |    |    |    |    |    |     |
|---------------------|-----|----|----|----|----|----|----|----|----|----|----|----|-----|
| Individuals ( $k$ ) | 1   | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 |     |
| Species ( $n_k$ )   | 118 | 74 | 44 | 24 | 29 | 22 | 20 | 19 | 20 | 15 | 12 | 14 |     |
| Individuals ( $k$ ) | 13  | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | Sum |
| Species ( $n_k$ )   | 6   | 12 | 6  | 9  | 9  | 6  | 10 | 10 | 11 | 5  | 3  | 3  | 5   |

```
data(Butterfly, package="vcd")  
barplot(Butterfly,  
        xlab = "Number of individuals",  
        ylab = "Number of species",  
        col = "pink",  
        cex.lab = 1.5)
```

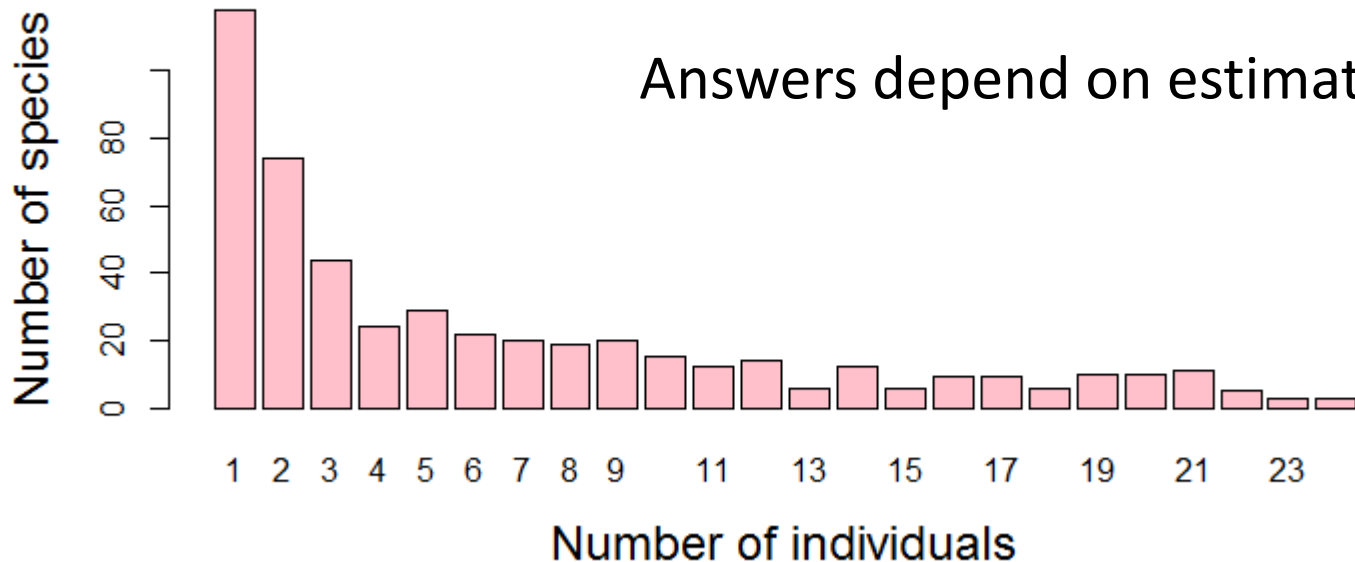
### Questions:

What is the total pop. of butterflies in Malaysia?

How many wolves remain in Canada NWT?

How many words did Shakespeare know?

Answers depend on estimating  $\Pr(k=0)$



# Discrete distributions: Questions

- General questions
  - What process gave rise to the distribution?
  - What is the form: uniform, binomial, Poisson, negative binomial, ... ?
  - → Fit & estimate parameters
    - Visualize goodness of fit
  - → Use in some larger context to tell a story
- Examples
  - *Families in Saxony*: might expect  $\text{Bin}(n=12, p)$ ;  $p=0.5$ ?
  - *HorseKicks*: Poisson ( $\lambda$ ); here,  $\lambda = \text{mean} = 0.61$
  - *Federalist papers*: Perhaps Poisson( $\lambda$ ) or NegBin ( $\lambda, p$ )
  - *Butterfly data*: Perhaps a log-series distribution?

# Fitting discrete distributions

## Lack of fit:

- Lack of fit tells us something about the *process* giving rise to the data
- Poisson: assumes constant small probability of the basic event
- Binomial: assumes constant probability and independent trials
- Negative binomial: allows for *overdispersion*, relative to Poisson

## Motivation:

- Models for more complex categorical data use these basic discrete distributions
- Binomial (with predictors) → logistic regression
- Poisson (with predictors) → poisson regression, loglinear models
- ⇒ many of these are special cases of *generalized linear models*

# Common discrete distributions

| Distribution            | Counts, $k$                            | Values of $X$    | $\Pr(X=k)$                          | Mean, $E(X)$       | Var, $V(X)$          |
|-------------------------|----------------------------------------|------------------|-------------------------------------|--------------------|----------------------|
| Bernoulli( $p$ )        | Success in 1 trial                     | $k=\{0, 1\}$     | $p^k(1-p)^{1-k}$                    | $p$                | $p(1-p)$             |
| Binomial( $n, p$ )      | # successes in $n$ trials              | $0, 1, \dots, n$ | $\binom{n}{k} p^k(1-p)^{n-k}$       | $np$               | $np(1-p)$            |
| Geometric( $p$ )        | # of trials to 1 <sup>st</sup> success | $0, 1, 2, \dots$ | $p(1-p)^k$                          | $\frac{1-p}{p}$    | $\frac{1-p}{p^2}$    |
| Neg. binomial( $k, p$ ) | # of trials to $k^{\text{th}}$ success | $0, 1, 2, \dots$ | $\binom{n+k-1}{k} p^k(1-p)^n$       | $\frac{k(1-p)}{p}$ | $\frac{k(1-p)}{p^2}$ |
| Poisson( $\lambda$ )    | # of events in interval                | $0, 1, 2, \dots$ | $\frac{\lambda^k e^{-\lambda}}{k!}$ | $\lambda$          | $\lambda$            |
| Log series( $p$ )       | # of types observed                    | $0, 1, 2, \dots$ | $\frac{p^k}{n \log(1-p)}$           |                    |                      |

# Discrete distributions: R

R functions: {d\_\_\_, p\_\_\_, q\_\_\_, r\_\_\_}

- d\_\_\_ **density** function,  $\Pr(X=k) = p(k)$
- p\_\_\_ **cumulative probability**,  $F(k) = \sum_{X \leq k} p(k)$
- q\_\_\_ **quantile** function, find  $k = F^{-1}(p)$ , smallest value such that  $F(k) \geq p$
- r\_\_\_ **random** number generator

| Discrete distribution | Density (pmf) function    | Cumulative (CDF)          | Quantile $CDF^{-1}$       | Random # generator        |
|-----------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| Binomial              | <code>dbinom()</code>     | <code>pbinom()</code>     | <code>qbinom()</code>     | <code>rbinom()</code>     |
| Poisson               | <code>dpois()</code>      | <code>ppois()</code>      | <code>qpois()</code>      | <code>rpois()</code>      |
| Negative binomial     | <code>dnbinom()</code>    | <code>pnbinom()</code>    | <code>qnbinom()</code>    | <code>rnbinom()</code>    |
| Geometric             | <code>dgeom()</code>      | <code>pgeom()</code>      | <code>qgeom()</code>      | <code>rgeom()</code>      |
| Logarithmic series    | <code>dlogseries()</code> | <code>plogseries()</code> | <code>qlogseries()</code> | <code>rlogseries()</code> |

e.g.,

```
> dbinom(0:4, size=4, p=1/2)      # number of H in 4 coin tosses
[1] 0.0625 0.2500 0.3750 0.2500 0.0625
> dpois(0:4, lambda=3)           # poisson, with  $\lambda = 3$ 
[1] 0.0498 0.1494 0.2240 0.2240 0.1680
```

# What is “binomial”

Bi-no-mi-al /bī'nōmēəl/

- **Taxonomy:** A **two-part name**, (genus, species) e.g., *Elephas maximus* for the Asian elephant
- **Mathematics:** An algebraic expression of a sum of **two terms**,  $(x + y)$  or expansion,  $(x + y)^n$

$$\begin{aligned}(x+y)^0 &= 1 \\(x+y)^1 &= 1x + 1y \\(x+y)^2 &= 1x^2 + 2x^1y^1 + 1y^2 \\(x+y)^3 &= 1x^3 + 3x^2y^1 + 3x^1y^2 + 1y^3 \\(x+y)^4 &= 1x^4 + 4x^3y^1 + 6x^2y^2 + 4x^1y^3 + 1y^4 \\(x+y)^5 &= 1x^5 + 5x^4y^1 + 10x^3y^2 + 10x^2y^3 + 5x^1y^4 + 1y^5\end{aligned}$$

Coefficients of terms

$$C_k^n = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

(Pascal's triangle)

# Binomial distribution

The binomial distribution,  $\text{Bin}(n, p)$ ,

$$\text{Bin}(n, p) : \Pr\{X = k\} \equiv p(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad k = 0, 1, \dots, n, \quad (1)$$

*Diagram annotations:*

- Red arrow from  $\binom{n}{k}$  to text: # ways to get k out of n
- Red arrow from  $p^k$  to text: Pr(k events)
- Red arrow from  $(1-p)^{n-k}$  to text: Pr(n-k non-events)

arises as the distribution of the number of events of interest (“successes”) which occur in  $n$  *independent trials* when the probability of the event on any one trial is the *constant* value  $p = \Pr(\text{event})$ .

## Examples

- Toss 10 fair coins– how many heads?  $\text{Bin}(10, \frac{1}{2})$
- Toss 12 fair dice– how many 5s or 6s?  $\text{Bin}(12, 1/3)$

Mean, variance, skewness:

$$\text{Mean}[X] = n p$$

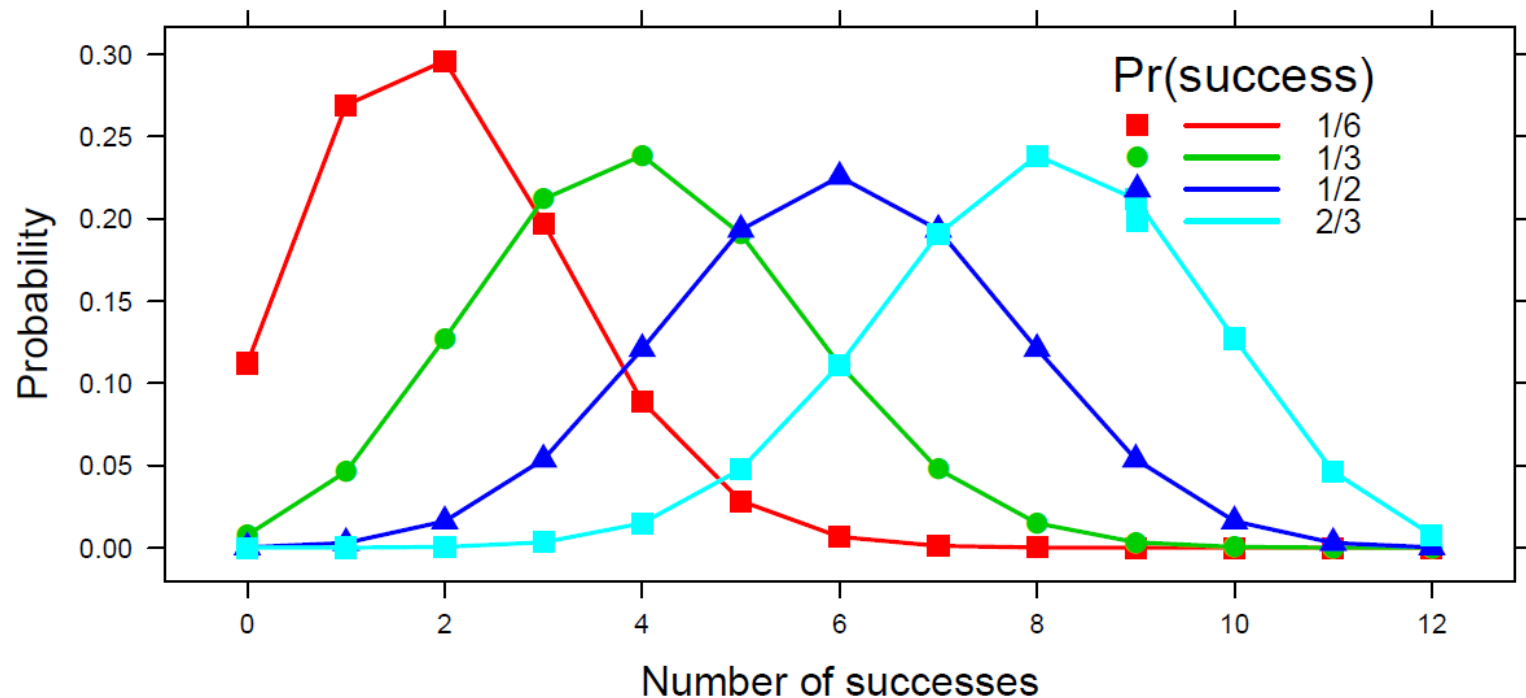
$$\text{Var}[X] = n p (1-p) = n p q$$

$$\text{Skew}[X] = n p q (q-p)$$

$$\text{MLE from data: } \hat{p} = \frac{\bar{x}}{n} = \frac{\sum_k k \times n_k / \sum_k n_k}{n}$$

# Binomial distribution

Binomial distributions for  $k = 0, 1, 2, \dots, 12$  successes in  $n=12$  trials, for 4 values of  $p$



- Mean =  $n p$
- Variance is maximum when  $p = \frac{1}{2}$
- Skewed when  $p \neq \frac{1}{2}$

# Poisson distribution

The Poisson distribution,  $\text{Pois}(\lambda)$ ,

$$\text{Pois}(\lambda) : \Pr\{X = k\} \equiv p(k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad k = 0, 1, \dots \quad (2)$$

gives the probability of an event occurring  $k = 0, 1, 2, \dots$  times over a *large number of independent* trials, when the probability,  $p$ , that the event occurs on any one trial (in time or space) is *small and constant*.

Examples:

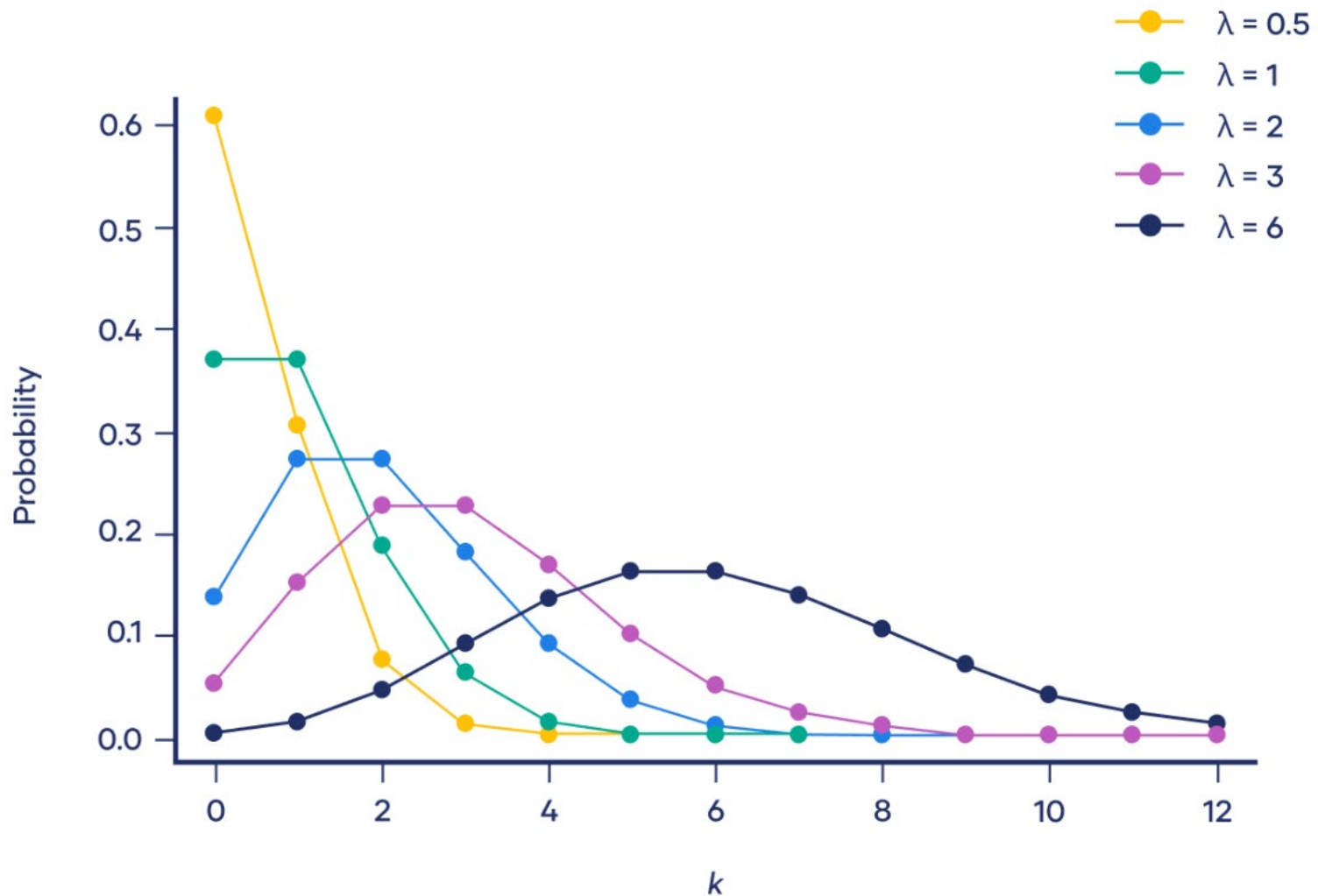
- Number of highway accidents at some given location
- Defects in a manufacturing process
- Number of goals scored in soccer games

**Table:** Total goals scored in 380 games in the Premier Football League, 1995/95 season

| Total goals     | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 |
|-----------------|----|----|----|----|----|----|----|---|
| Number of games | 27 | 88 | 91 | 73 | 49 | 31 | 18 | 3 |

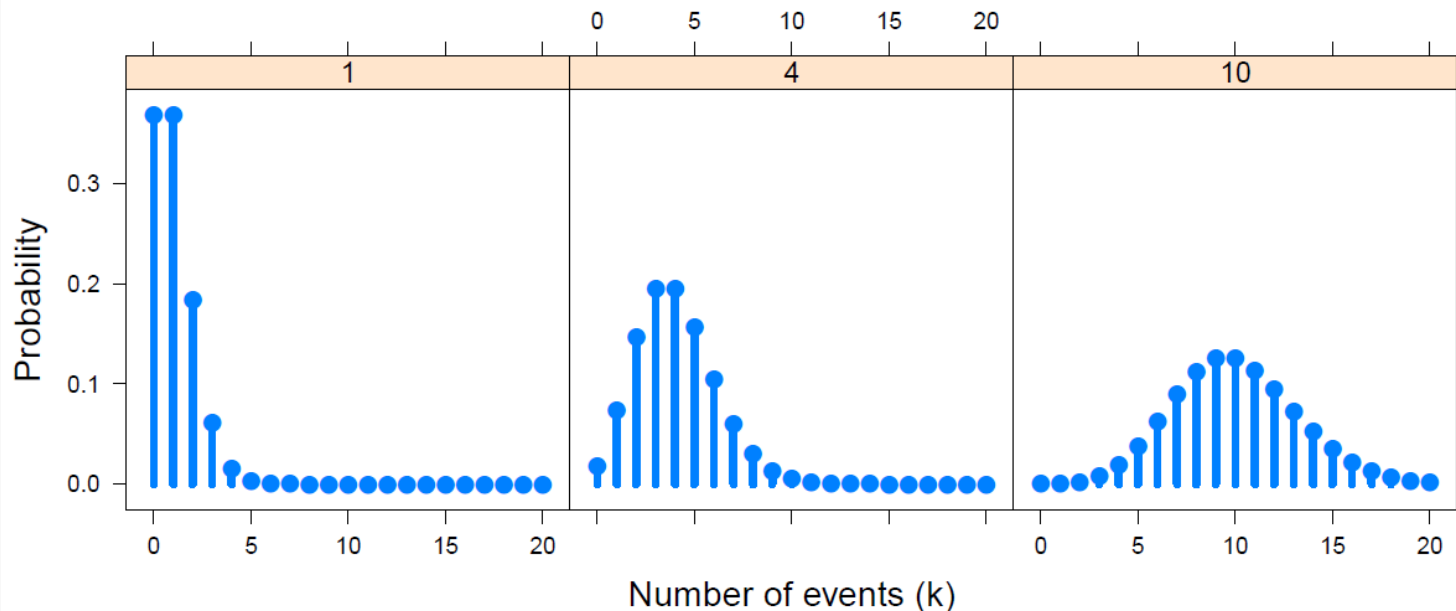
# Poisson distribution

Poisson distributions for  $\lambda = \frac{1}{2}, 1, 2, 3, 6$



# Poisson distribution: Properties

Poisson distributions for  $\lambda = 1, 4, 10$



DDAR Fig  
3.10, p 81

**Mean, variance, skewness:**

$$\text{Mean}[X] = \lambda$$

$$\text{Var}[X] = \lambda$$

$$\text{Skew}[X] = \lambda^{-1/2}$$

$$\text{MLE: } \hat{\lambda} = \bar{x}$$

**Properties:**

Sum of Pois ( $\lambda_1, \lambda_2, \lambda_3, \dots$ ) = Pois( $\sum \lambda_i$ )

Approaches  $N(\lambda, \lambda)$  as  $n \rightarrow \infty$

# History: Who discovered the “Poisson” distribution

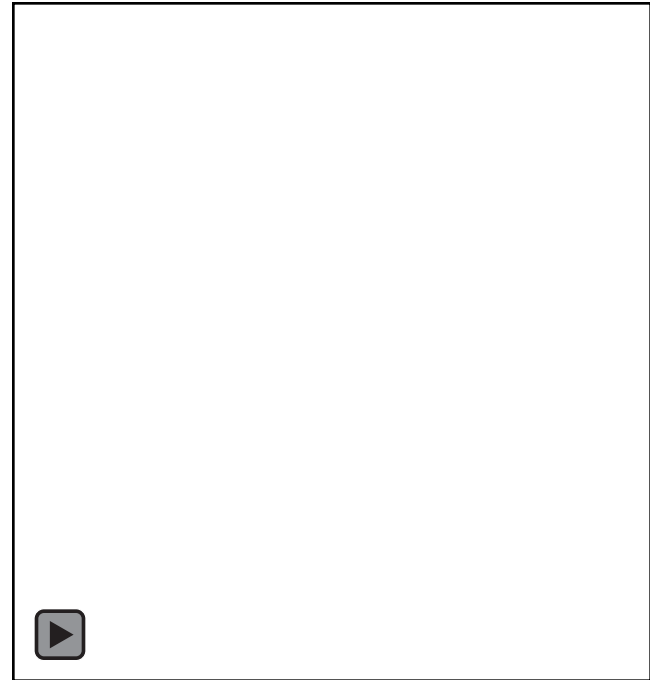
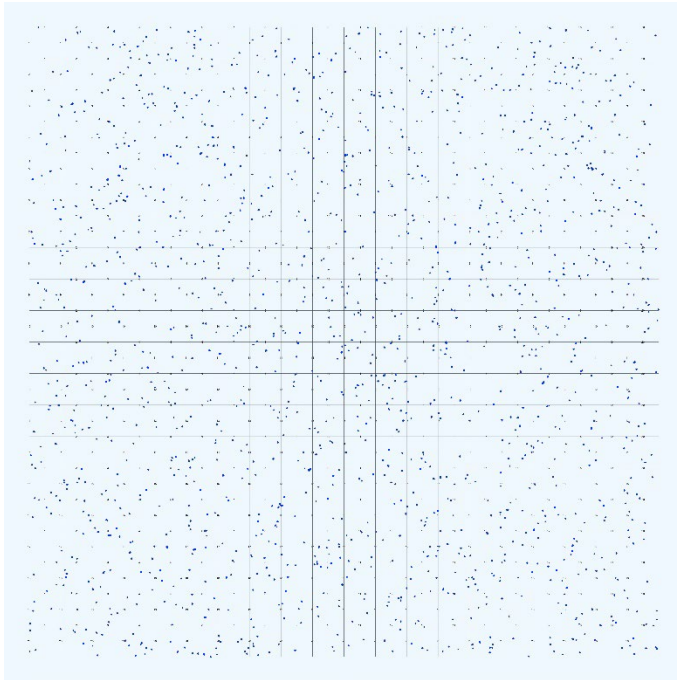
**Stigler’s Law:** No discovery in science is ever named for its primary originator

- De Moivre (1718) – Approximation to binomial as  $n$  gets largish
- **Poisson (1837)** – *Reserches sur la Probabilité des jugements en Matière criminelle...* -- Derives  $e^{-\lambda} \lambda^k / k!$ 
  - Stigler says main result anticipated by De Moivre
- S. Newcomb (1860) – *Notes on a Theory of Probability*
  - First attempt at using this as a fit to **data**
  - Observations of stars:  $\Pr(\text{any small space, } 1^\circ) \text{ contains } s \text{ stars, } s = 0, 1, 2, \dots$
- Von Bortkiewicz (1898) – *Law of Small Numbers*
  - Re-derives Poisson as limiting case of binomial
  - Several data sets (Horse kicks & others) – “agreement between theory and observation leave nothing to be desired”
- Gosset (1907): Heamacytometer Counts
  - “Student”’s first paper – first rigorous treatment of the Poisson for count data

See: Hanley & Bhatnagar (2022) The “Poisson” Distribution: History, Reenactments, Adaptations, *The American Statistician*, 76:4, 363-371, [DOI: 10.1080/00031305.2022.2046159](https://doi.org/10.1080/00031305.2022.2046159)

# Gosset: Heamacytometer Counts

Number of blood cells observed in a 20 x 20 grid on a slide



Source: <http://www.medicine.mcgill.ca/epidemiology/hanley/Gosset/>

# Negative binomial distribution

The Negative binomial distribution,  $\text{NBin}(n, p)$ ,

$$\text{NBin}(n, p) : \Pr\{X = k\} \equiv p(k) = \binom{n+k-1}{k} p^n (1-p)^k \quad k = 0, 1, \dots, \infty$$

arises when a series of independent Bernoulli trials is observed with constant probability  $p$  of some event, and we ask how many non-events (failures),  $k$ , it takes to observe  $n$  successful events.

Example: Toss a coin; what is probability of getting  $k = 0, 1, 2, \dots$  tails before  $n = 3$  heads?

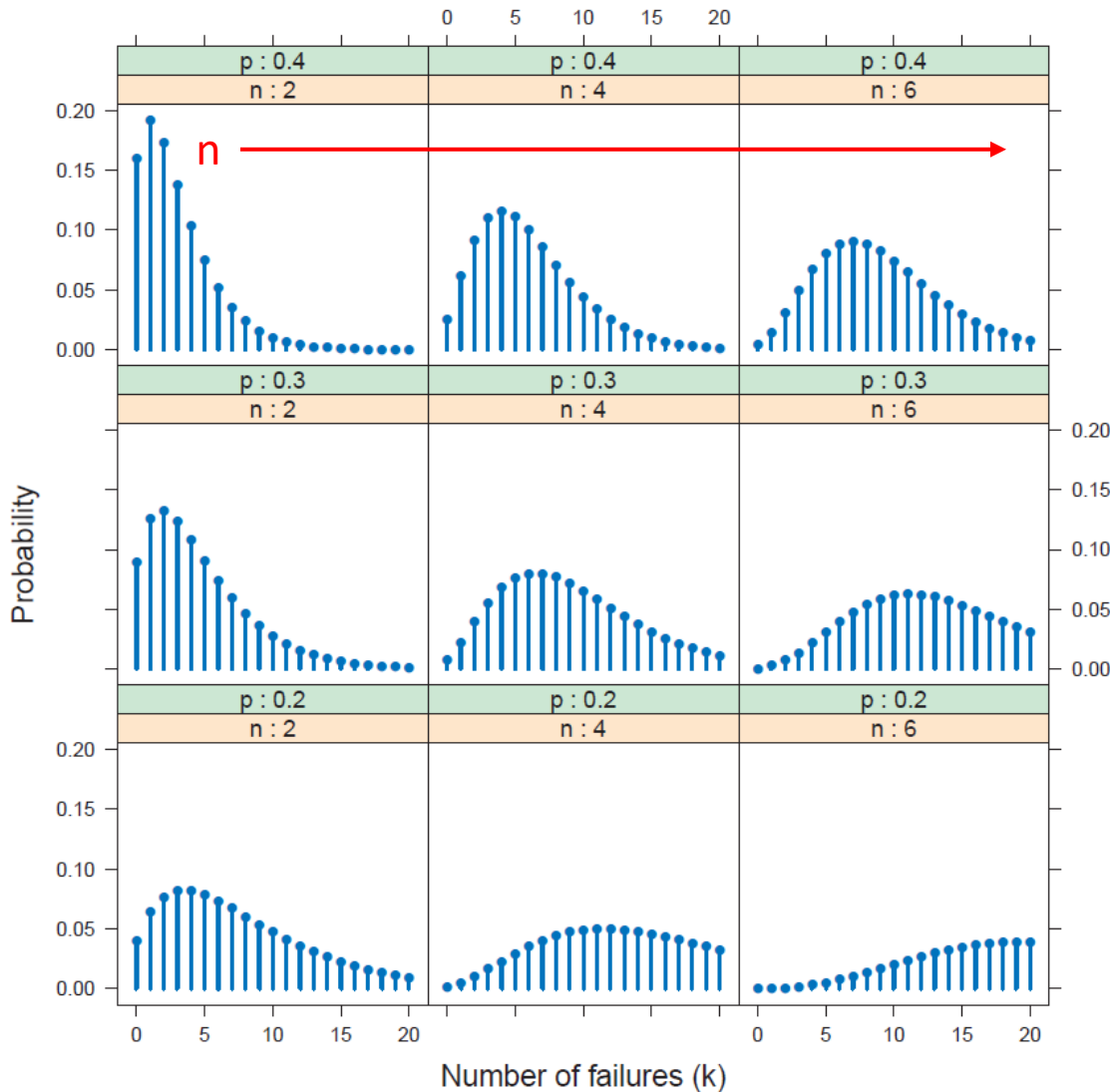
This distribution is often used as an alternative to the Poisson when

- constant probability  $p$  or independence are violated
- variance is greater than the mean (overdispersion)

Properties:

$$\text{Mean}(X) = \mu = \frac{n(1-p)}{p} \implies p = \frac{n}{n+\mu},$$

$$\text{Var}(X) = \frac{n(1-p)}{p^2} \implies \boxed{\text{Var}(X) = \mu + \frac{\mu^2}{n}}.$$



Negative binomial  
distributions for  
 $n = 2, 4, 6$   
 $p = 0.2, 0.3, 0.4$

Mean:  
Increases with  $n$   
Decreases with  $p$

DDAR Fig 3.13, p 85

# Quiz: Name that distribution

1. Weldon tossed 12 dice 26,306 times & tallied the number of times a 5 or 6 occurred

```
> WeldonDice
```

```
n56
```

| 0   | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8   | 9   | 10+ |
|-----|------|------|------|------|------|------|------|-----|-----|-----|
| 185 | 1149 | 3265 | 5475 | 6114 | 5194 | 3067 | 1331 | 403 | 105 | 18  |

**Bin(n=12, p=1/3)**

2. Pele practices penalty kicks for the upcoming 1958 FIFA World Cup. His average scoring has been  $p=0.4$ . What is the probability it will take him 1, 2, ... shots to score a goal?

**Nbin(n=1, p=0.4)**

```
> dnbinom(1:5, size=1, p=0.4)
[1] 0.240 0.144 0.086 0.052 0.031
```

3. A Geiger counter records the number of scintillations of  $\alpha$  particles from a radioactive source, with an average rate of 20/msec. What is the probability of observing 40 in a 1 msec. interval?

**Pois( $\lambda=20$ )**

4. What is the distribution of the time between Geiger counter ticks?

**Exponential dist<sup>n</sup>,  $\Pr(X=k) = \lambda e^{-\lambda k}$ , mean =  $1/\lambda$**

# Fitting discrete distributions

Fitting a discrete distribution involves the following steps:

- 1 Estimate the parameter(s) from the data, e.g.,  $p$  for binomial,  $\lambda$  for Poisson, etc. Typically done using maximum likelihood, but some distributions have simple expressions:
  - Binomial,  $\hat{p} = \sum kn_k / (n \sum n_k) = \text{mean} / n$
  - Poisson,  $\hat{\lambda} = \sum kn_k / \sum n_k = \text{mean}$
- 2 Calculate fitted probabilities,  $\hat{p}(k)$  for the distribution, and then fitted frequencies,  $N\hat{p}(k)$ .
- 3 Assess Goodness of fit: Pearson  $\chi^2$  or likelihood-ratio  $G^2$

$$\chi^2 = \sum_{k=1}^K \frac{(n_k - N\hat{p}_k)^2}{N\hat{p}_k} \quad G^2 = \sum_{k=1}^K n_k \log\left(\frac{n_k}{N\hat{p}_k}\right)$$

Both have asymptotic chisquare distributions,  $\chi^2_{K-s}$  with  $s$  estimated parameters, under the hypothesis that the data follows the chosen distribution.

# Fitting: Weldon's dice

Basic, naïve calculation of expected frequencies for a binomial distribution

```
> data(WeldonDice, package="vcd")
> Weldon.df <- as.data.frame(WeldonDice)      # convert to data frame

> Prob <- dbinom(0:12, 12, 1/3)                # binomial probabilities
> Prob <- c(Prob[1:10], sum(Prob[11:13]))      # sum values for 10+
> Exp = round(sum(WeldonDice)*Prob)           # expected frequencies
> Diff = Weldon.df[, "Freq"] - Exp             # raw residuals
> Chisq = Diff^2 / Exp                         # contribution to chisquare
> data.frame(Weldon.df, Prob=round(Prob,5), Exp, Diff, Chisq)
```

|    | n56 | Freq | Prob    | Exp  | Diff | Chisq |
|----|-----|------|---------|------|------|-------|
| 1  | 0   | 185  | 0.00771 | 203  | -18  | 1.596 |
| 2  | 1   | 1149 | 0.04624 | 1216 | -67  | 3.692 |
| 3  | 2   | 3265 | 0.12717 | 3345 | -80  | 1.913 |
| 4  | 3   | 5475 | 0.21195 | 5576 | -101 | 1.829 |
| 5  | 4   | 6114 | 0.23845 | 6273 | -159 | 4.030 |
| 6  | 5   | 5194 | 0.19076 | 5018 | 176  | 6.173 |
| 7  | 6   | 3067 | 0.11127 | 2927 | 140  | 6.696 |
| 8  | 7   | 1331 | 0.04769 | 1255 | 76   | 4.602 |
| 9  | 8   | 403  | 0.01490 | 392  | 11   | 0.309 |
| 10 | 9   | 105  | 0.00331 | 87   | 18   | 3.724 |
| 11 | 10+ | 18   | 0.00054 | 14   | 4    | 1.143 |

Doesn't calculate the MLE,  $\hat{p}$   
Manually sum  $k \geq 10$

# Fitting & graphing discrete distributions

In R, the **vcd** and **vcdExtra** packages provide functions to fit, visualize and diagnose discrete distributions

- **Fitting:** `goodfit()` fits uniform, binomial, Poisson, neg bin, geometric, logseries, ...
- **Graphing:** `rootogram()` assess departure between observed, fitted counts
- **Ord plot:** `Ordplot()` diagnose form of a discrete distribution
- **Robust plots:** `distplot()` handle problems with discrepant counts

# Example: Saxony families

```
> data(Saxony, package="vcd")
```

```
> Saxony
```

```
nMales
```

|   |    |     |     |     |      |      |      |     |     |     |    |    |
|---|----|-----|-----|-----|------|------|------|-----|-----|-----|----|----|
| 0 | 1  | 2   | 3   | 4   | 5    | 6    | 7    | 8   | 9   | 10  | 11 | 12 |
| 3 | 24 | 104 | 286 | 670 | 1033 | 1343 | 1112 | 829 | 478 | 181 | 45 | 7  |

Use **goodfit()** to fit the binomial; test with **summary()**

```
> Sax.fit <- goodfit(Saxony, type = "binomial", par=list(size=12))
```

```
> summary(Sax.fit)
```

Goodness-of-fit test for binomial distribution

|                  | X^2 | df | P(> X^2) |
|------------------|-----|----|----------|
| Likelihood Ratio | 97  | 11 | 6.98e-16 |

Doesn't  
fit. Why?

Specify parameters

# Example: Saxony families

The `print()` method for **goodfit** objects shows the details

```
> Sax.fit      # print
```

Observed and fitted values for binomial distribution  
with parameters estimated by 'ML'

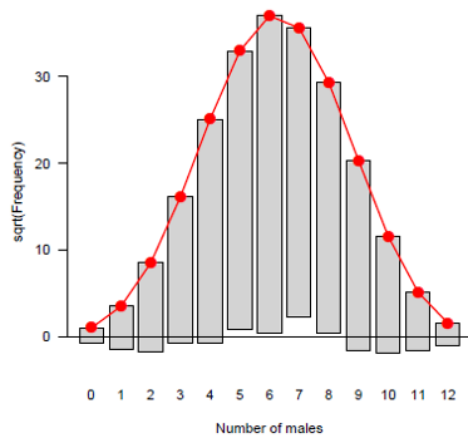
| count | observed | fitted   | pearson | residual |
|-------|----------|----------|---------|----------|
| 0     | 3        | 0.933    |         | 2.140    |
| 1     | 24       | 12.089   |         | 3.426    |
| 2     | 104      | 71.803   |         | 3.800    |
| 3     | 286      | 258.475  |         | 1.712    |
| 4     | 670      | 628.055  |         | 1.674    |
| 5     | 1033     | 1085.211 |         | -1.585   |
| 6     | 1343     | 1367.279 |         | -0.657   |
| 7     | 1112     | 1265.630 |         | -4.318   |
| 8     | 829      | 854.247  |         | -0.864   |
| 9     | 478      | 410.013  |         | 3.358    |
| 10    | 181      | 132.836  |         | 4.179    |
| 11    | 45       | 26.082   |         | 3.704    |
| 12    | 7        | 2.347    |         | 3.037    |

Pay attention to the  
**signs & magnitudes** of  
residuals,  $d_k$

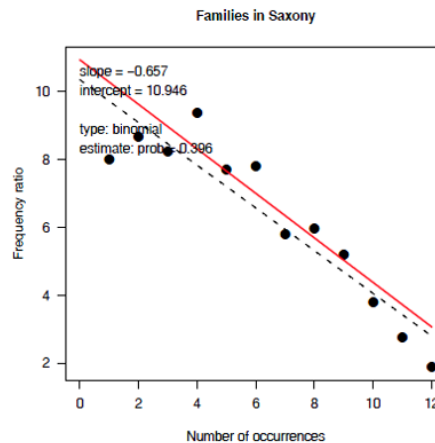
$$\text{Pearson } \chi^2 = \sum d_k^2$$

# Graphing discrete distributions

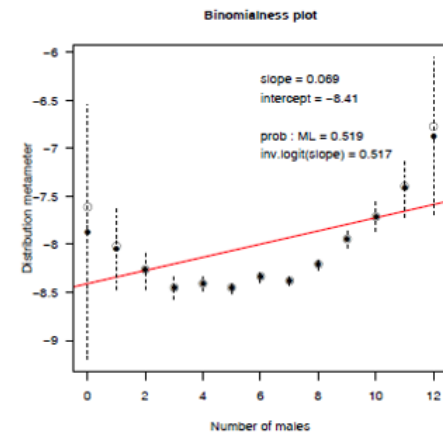
Rootograms



Ord plots



Robust  
distribution  
plots

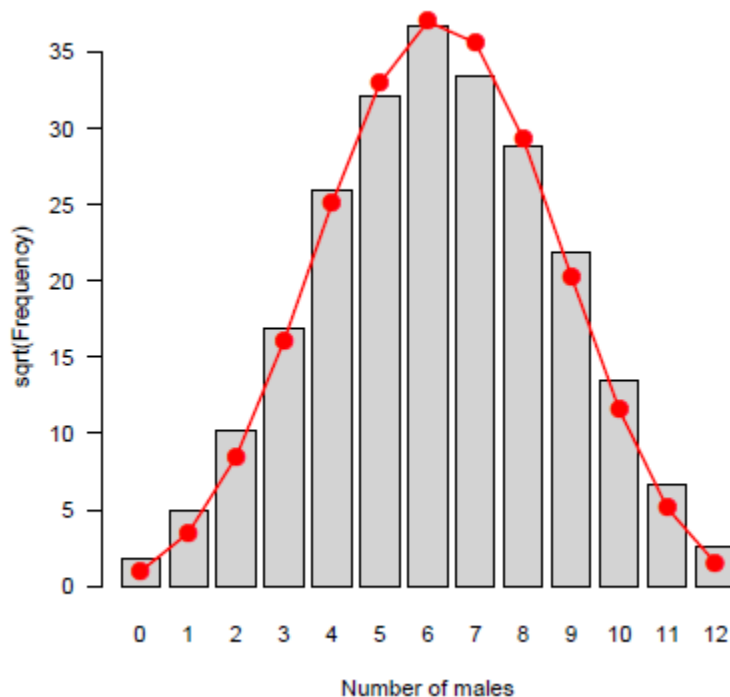


# What's wrong with simple histograms?

Discrete distributions are often graphed as histograms, with a theoretical fitted distribution superimposed

The plot() method for goodfit objects provides some alternatives

```
> plot(Sax.fit, type = "standing", xlab = "Number of males")
```

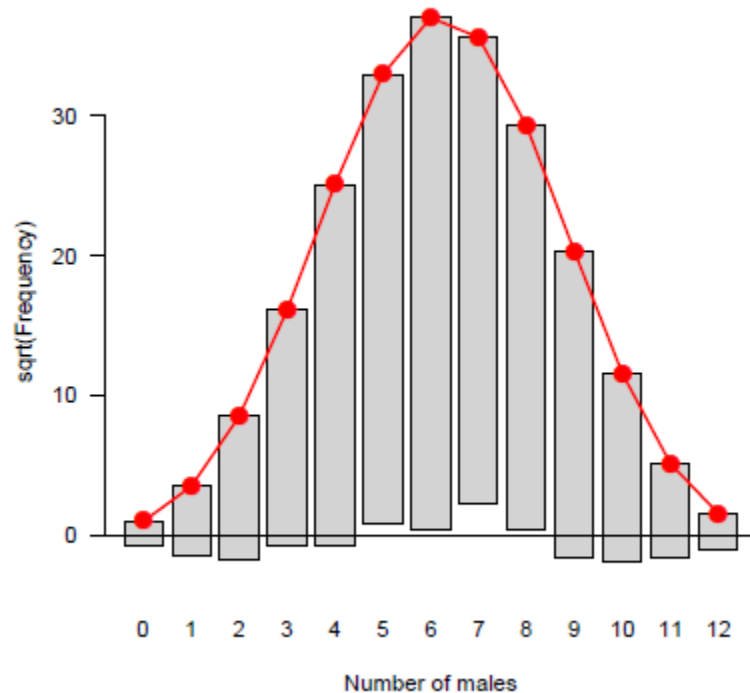


Problems:

- Largest frequencies dominate
- Must assess deviations vs. the fitted curve

# Hanging rootograms

```
> plot(Sax.fit, type = "hanging", xlab = "Number of males") # default
```



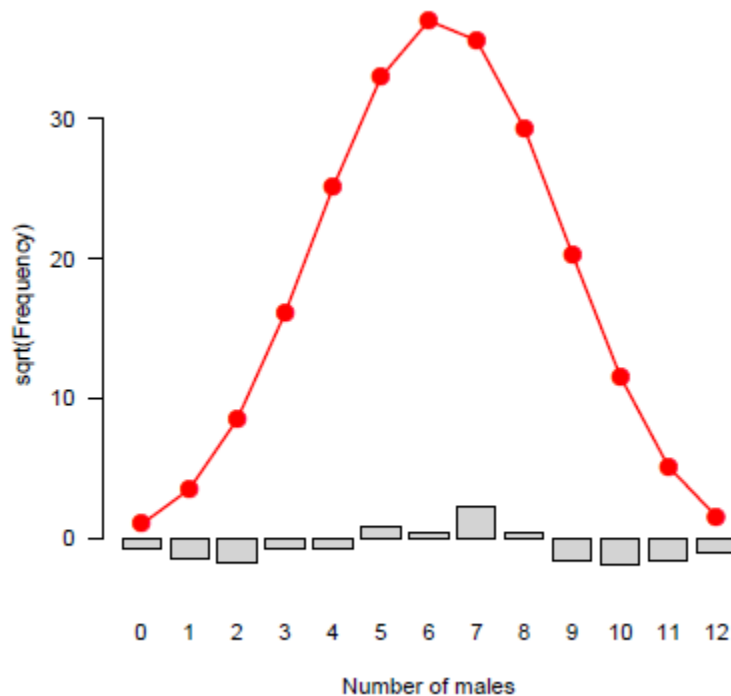
Tukey (1972, 1977):

- shift histogram bars to the fitted curve
- → judge deviations vs. horizontal line.
- plot  $\sqrt{\text{freq}}$  → smaller frequencies are emphasized.

We can now see clearly *where* the binomial doesn't fit

# Deviation rootograms

```
> plot(Sax.fit, type = "deviation", xlab = "Number of males")
```



Deviation rootogram:

- emphasize differences between observed and fitted frequencies
- bars now show the residuals (gaps) directly

There are more families with very low or very high number of sons than the binomial predicts.

Q: Why is this so much better than the lack-of-fit test?

Q: What does this suggest about sex distribution of families in Saxony?

# Example: Federalist papers

```
> data(Federalist, package="vcd")
```

```
> Federalist
```

```
nMay
```

|     |    |    |   |   |   |   |
|-----|----|----|---|---|---|---|
| 0   | 1  | 2  | 3 | 4 | 5 | 6 |
| 156 | 63 | 29 | 8 | 4 | 1 | 1 |

## Fit the Poisson distribution

```
> Fed.fit0 <- goodfit(Federalist, type="poisson")
```

```
> summary(Fed.fit0)
```

```
Goodness-of-fit test for poisson  
distribution
```

|                  | X^2  | df | P(> X^2) |
|------------------|------|----|----------|
| Likelihood Ratio | 25.2 | 5  | 0.000125 |

This fits very poorly!

# Example: Federalist papers

Try the Negative binomial distribution

```
> Fed.fit1<- goodfit(Federalist, type="nbinomial")  
> summary(Fed.fit1)
```

Goodness-of-fit test for nbinomial distribution

|                  | X^2  | df | P(> X^2) |
|------------------|------|----|----------|
| Likelihood Ratio | 1.96 | 4  | 0.742    |

This now fits very well, indeed! Why?

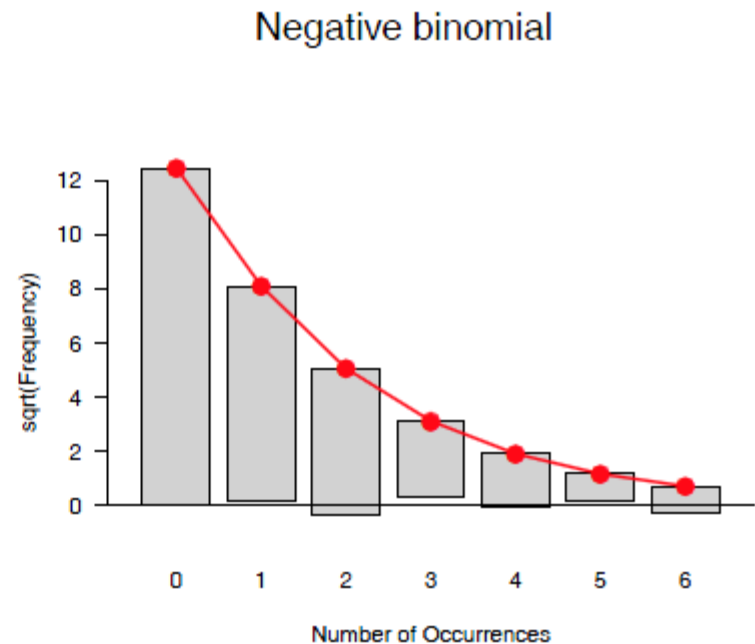
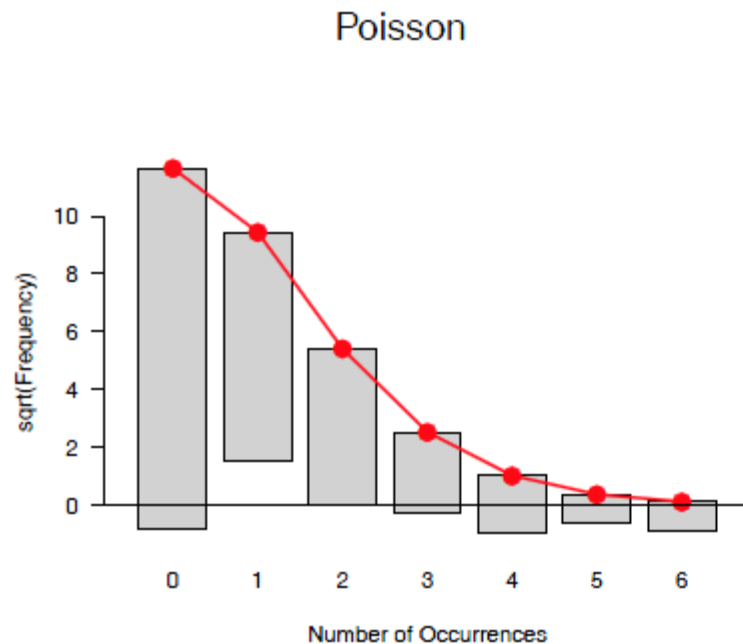
- Poisson assumes that the probability of a given word (“may”) is constant across all blocks of text.
- Negative binomial allows the rate parameter  $\lambda$  to vary over blocks of text

In their landmark study, Mosteller & Wallace used a number of such words to distinguish between papers written by Madison & Hamilton and then assign the “unknown” author papers --- all to James Madison

# Federalist papers: Rootograms

Hanging rootograms for the Federalist papers data, comparing Poisson and Negative binomial

```
> plot(Fed.fit0, main = "Poisson")  
> plot(Fed.fit1, main = "Negative binomial")
```

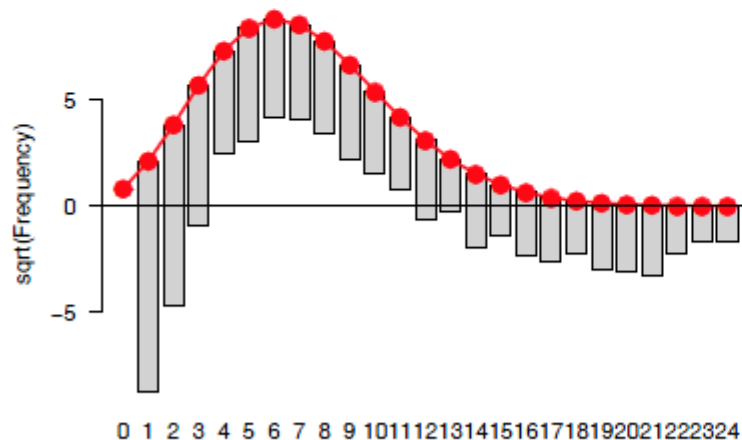


# Butterfly data

Both Poisson and Negative binomial are terrible fits! What to do??

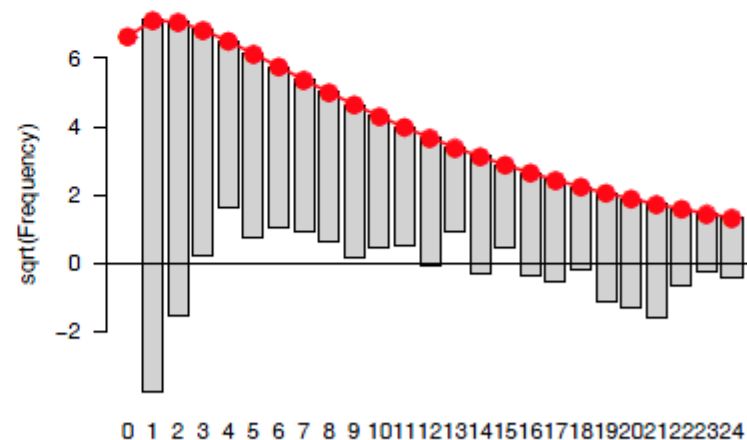
```
But.fit1 <- goodfit(Butterfly, type="poisson")  
But.fit2 <- goodfit(Butterfly, type="nbinomial")  
plot(But.fit1, main="Poisson")  
plot(But.fit2, main="Negative binomial")
```

Poisson



Number of Occurrences

Negative binomial



Number of Occurrences

# Ord plots: Diagnose form of distribution

How to tell which discrete distributions are likely candidates?

- Ord (1967): for each of Poisson, Binomial, Negative binomial, and Logarithmic series distributions,

- plot of  $kp_k/p_{k-1}$  against  $k$  is linear
- signs of intercept and slope  $\rightarrow$  determine the form, give rough estimates of parameters

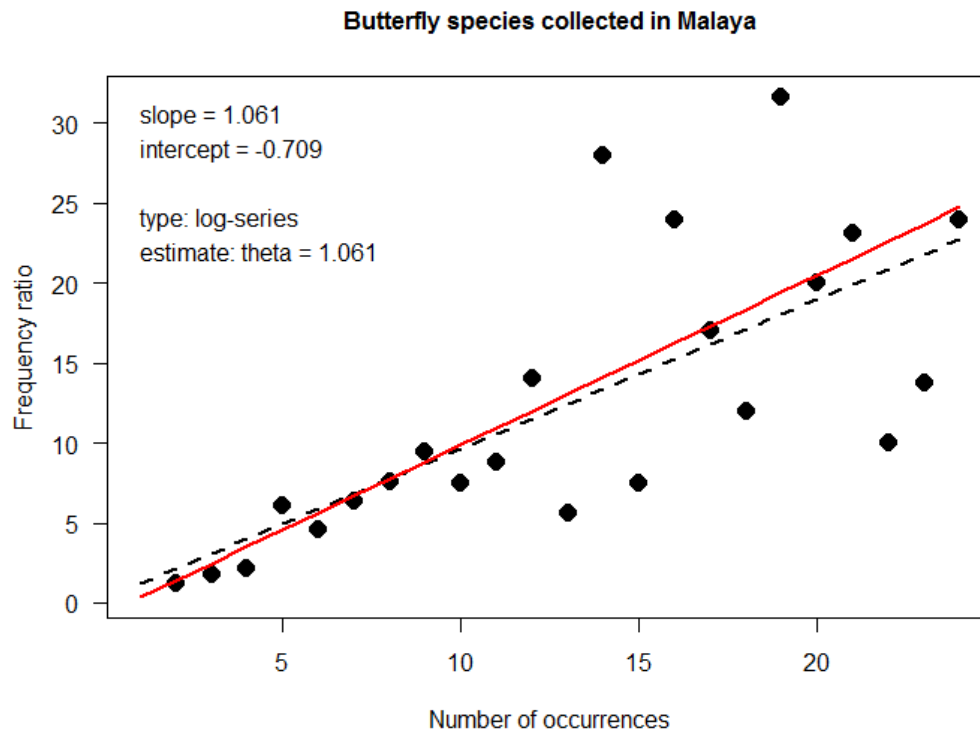
| Slope<br>(b) | Intercept<br>(a) | Distribution<br>(parameter) | Parameter<br>estimate         |
|--------------|------------------|-----------------------------|-------------------------------|
| 0            | +                | Poisson ( $\lambda$ )       | $\lambda = a$                 |
| —            | +                | Binomial ( $n, p$ )         | $p = b/(b - 1)$               |
| +            | +                | Neg. binomial ( $n, p$ )    | $p = 1 - b$                   |
| +            | —                | Log. series ( $\theta$ )    | $\theta = b$<br>$\theta = -a$ |

- Fit line by WLS, using  $\sqrt{n_k - 1}$  as weights
- A heuristic method: doesn't always work, but often a good start.

# Ord plot: Examples

**Butterfly data:** The slope and intercept correctly diagnoses the **log-series** distribution

```
> Ord_plot(Butterfly,  
            main = "Butterfly species collected in Malaya",  
            gp=gpar(cex=1), pch=16)
```



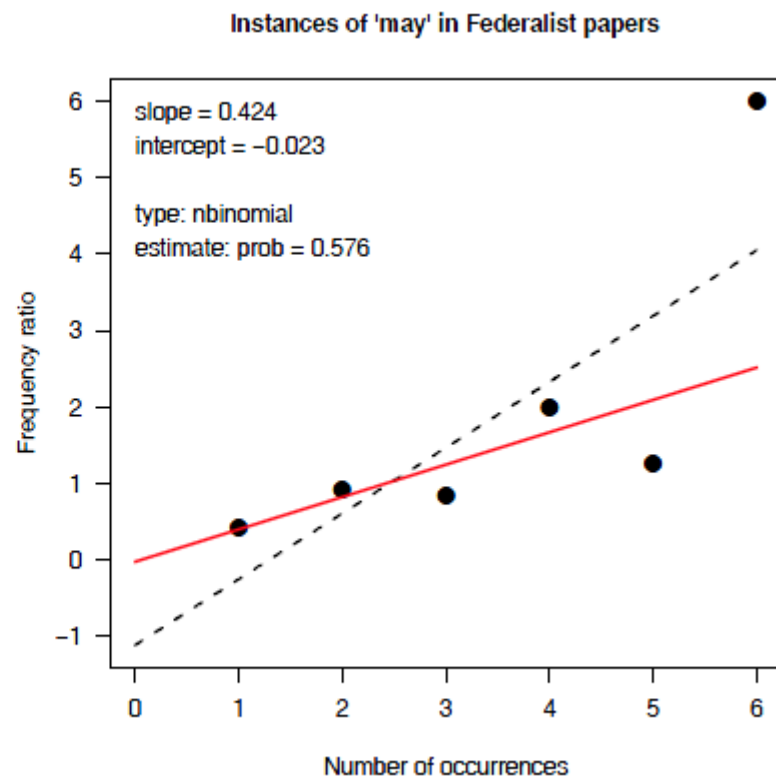
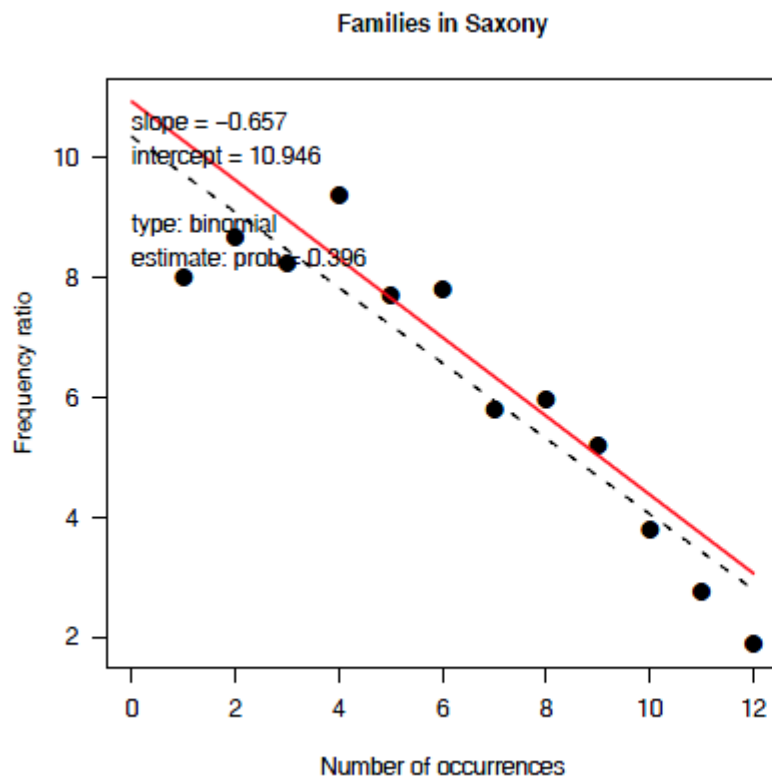
+ slope  
- intercept  
→ **log-series**

OLS line shown in black  
WLS line shown in red

# Ord plots: Examples

## Ord plots for the Saxony and Federalist data

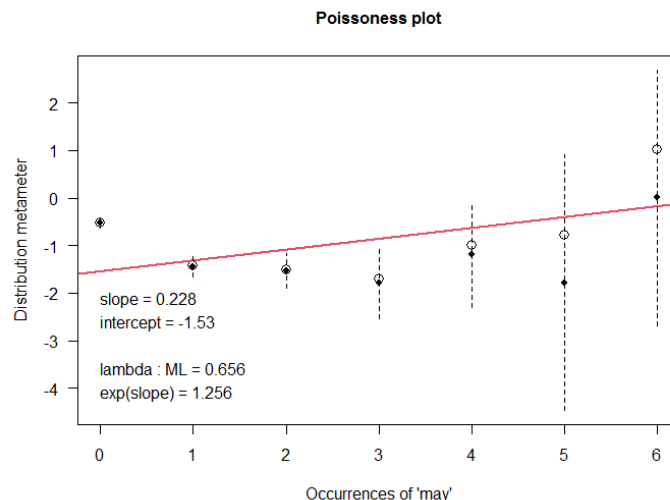
```
> Ord_plot(Saxony, main = "Families in Saxony", gp=gpar(cex=1), pch=16)  
> Ord_plot(Federalist, main = "Instances of 'may' in Federalist papers", gp=gpar(cex=1), pch=16)
```



# Robust distribution plots

- Ord plots lack robustness
  - one discrepant frequency,  $n_k$  affects points for both  $k$  and  $k + 1$
  - the use of WLS to fit the line is a small attempt to minimize this
- Robust plots for Poisson distribution (Hoaglin and Tukey, 1985)
  - For Poisson, plot **count metameter** =  $\phi(n_k) = \log_e(k! n_k / N)$  vs.  $k$
  - Linear relation  $\Rightarrow$  Poisson, slope gives  $\hat{\lambda}$
  - CI for points, diagnostic (influence) plot
  - Implemented in `distplot()` in the `vcd` package

For the Poisson distribution, this is called a “poissonness plot”



# Poissonness plot: Details

- If the distribution of  $n_k$  is  $\text{Poisson}(\lambda)$  for some fixed  $\lambda$ , then each observed frequency,  $n_k \approx m_k = Np_k$ .
- Then, setting  $n_k = Np_k = e^{-\lambda} \lambda^k / k!$ , and taking logs of both sides gives

$$\log(n_k) = \log N - \lambda + k \log \lambda - \log k!$$

which can be rearranged to

$$\phi(n_k) \equiv \log \left( \frac{k! n_k}{N} \right) = -\lambda + (\log \lambda) k$$

- $\Rightarrow$  if the distribution is Poisson, plotting  $\phi(n_k)$  vs.  $k$  should give a line with
  - intercept =  $-\lambda$
  - slope =  $\log \lambda$
- Nonlinear relation  $\rightarrow$  distribution is *not* Poisson
- Hoaglin and Tukey (1985) give details on calculation of confidence intervals and influence measures.

# Other distributions

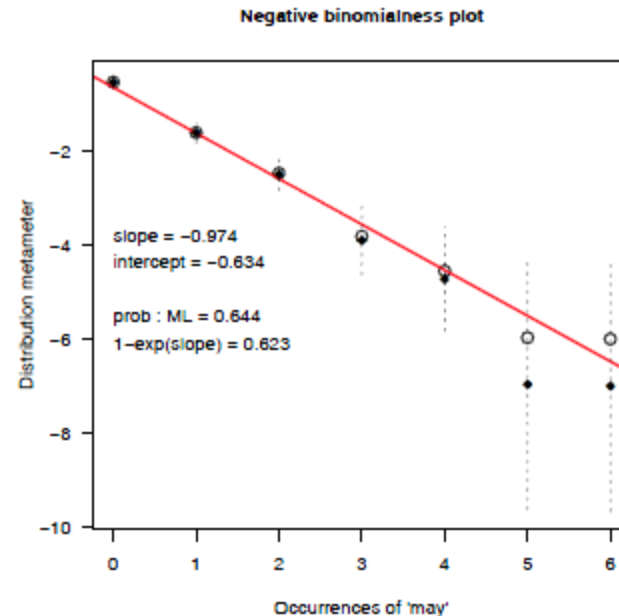
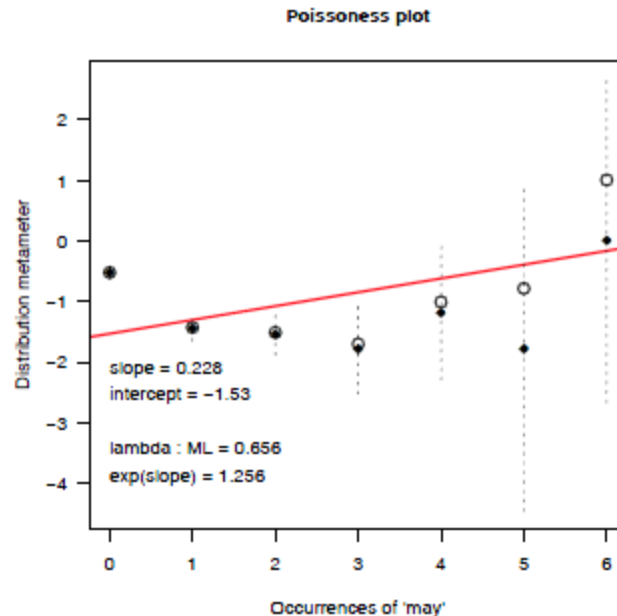
This idea extends readily to other discrete data distributions:

- The binomial, Poisson, negative binomial, geometric and logseries distributions are all members of a general **power series family** of discrete distributions. See: *DDAR*, Table 3.10 for details.
- This allows all of these to be represented in a plot of a suitable count metameter,  $\phi(n_k)$  vs.  $k$ . See: *DDAR*, Table 3.12 for details.
- In these plots, a straight line confirms that the data follow the given distribution.
- Confidence intervals around the points indicate **uncertainty** for the count metameter.
- The slope and intercept of the line give **estimates** of the distribution parameters.

# distplot: Federalist

Try both Poisson & Negative binomial

```
distplot(Federalist, type="poisson", xlab="Occurrences of 'may'")  
distplot(Federalist, type="nbinomial", xlab="Occurrences of 'may'")
```

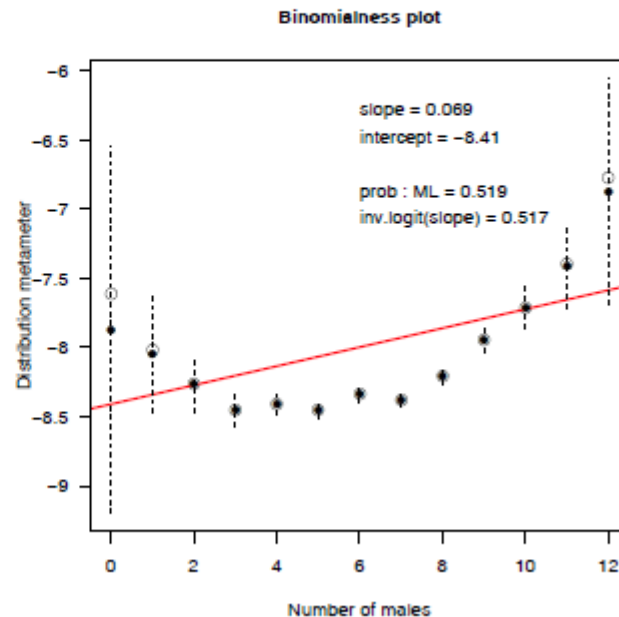
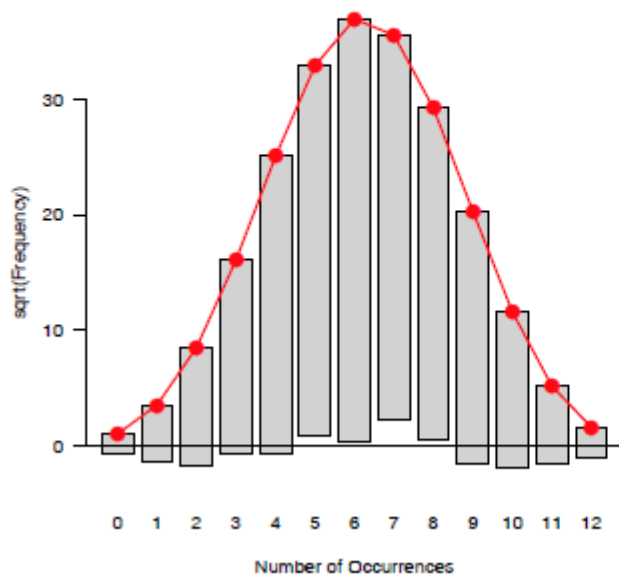


Again, the Poisson distribution is seen not to fit, while the Negative binomial appears reasonable.

# distplot: Saxony

For purported binomial distributions, the result is a “binomialness” plot

```
plot(goodfit(Saxony, type="binomial", par=list(size=12)))  
distplot(Saxony, type="binomial", size=12, xlab="Number of males")
```



Both plots show heavier tails than the binomial distribution. `distplot()` is more sensitive in diagnosing this

# What have we learned?

Main points:

- Discrete distributions involve basic *counts* of occurrences of some event occurring with varying *frequency*.
- The ideas and methods for one-way tables are building blocks for analysis of more complex data.
- Commonly used discrete distributions include the binomial, Poisson, negative binomial, and logarithmic series distributions, all members of a *power series* family.
- Fitting observed data to a distribution  $\rightarrow$  fitted frequencies,  $N\hat{p}_k$ ,  $\rightarrow$  goodness-of-fit tests (Pearson  $X^2$ , LR  $G^2$ )
- R: `goodfit()` provides `print()`, `summary()` and `plot()` methods.
- Plotting with rootograms, Ord plots and generalized distribution plots can reveal *how* or *where* a distribution does not fit.

# What have we learned?

Some explanations:

- The Saxony data were part of a much larger data set from Geissler (1889) (Geissler in [vcdExtra](#)).
  - For the binomial, with families of size  $n = 12$ , our analyses give  $\hat{p} = \Pr(\text{male}) = 0.52$ .
  - Other analyses (using more complex models) conclude that  $p$  varies among families with the same size.
  - One explanation is that family decisions to have another child are influenced by the boy–girl ratio in earlier children.
- As suggested earlier, the lack of fit of the Poisson distribution for words in the Federalist papers can be explained by [context](#) of the writing:
  - Given “marker” words appear more or less often over time and subject than predicted by constant rates ( $\lambda$ ) for a given author (Madison or Hamilton)
  - The negative binomial distribution fit much better.
  - The estimated parameters for these texts allowed assigning all 12 disputed papers to Madison.

# Looking ahead: PhdPubs data

Example 3.24 in DDAR gives data on the number of publications by PhD candidates in the last 3 years of study

```
data("PhdPubs", package = "vcdExtra")  
table(PhdPubs$articles)
```

```
##  
##      0      1      2      3      4      5      6      7      8      9     10     11     12     16     19  
## 275 246 178  84  67  27  17  12   1   2   1   1   2   1   1
```

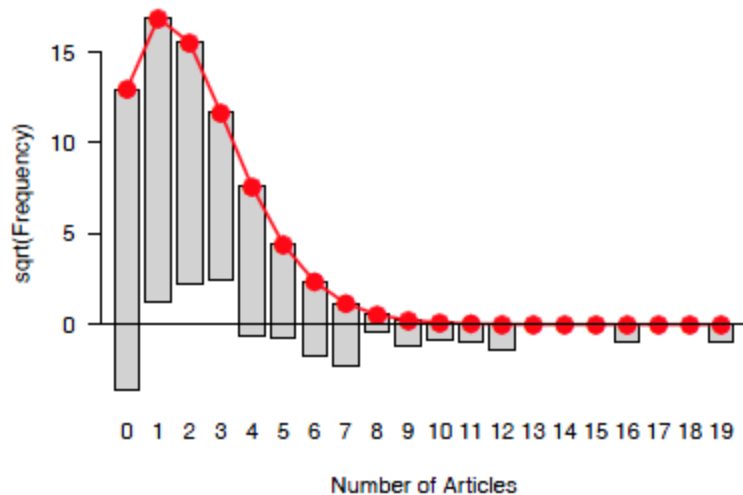
N = 915, mean(articles) = 1.69

- There are **predictors**: gender, marital status, number of children, prestige of dept., # pubs by student's mentor
- We fit such models with **glm()**, but need to specify the **form** of the distribution
- Ignoring the predictors for now, a baseline model could be  
`glm(articles ~ 1, data=PhdPubs, family = "poisson")`

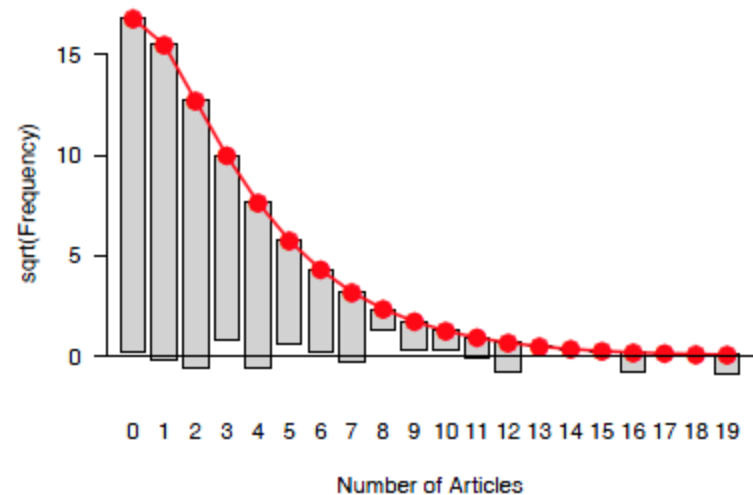
# Looking ahead: PhdPubs

```
plot(goodfit(PhdPubs$articles), xlab = "Number of Articles",  
     main = "Poisson")  
plot(goodfit(PhdPubs$articles, type = "nbinomial"),  
     xlab = "Number of Articles", main = "Negative binomial")
```

Poisson



Negative binomial



Poisson doesn't fit: Need to account for **excess 0s** (some never published)

Neg binomial: Sort of OK, but should take predictors into account

# Looking ahead: Count data models

Count data regression models (DDAR Ch 11)

- Include predictors
- Allow different distributions for **unexplained** variation
- Provide tests of one model vs. another
- Special models handle the problems of **excess zeros**: `zeroinfl()`, `hurdle()`

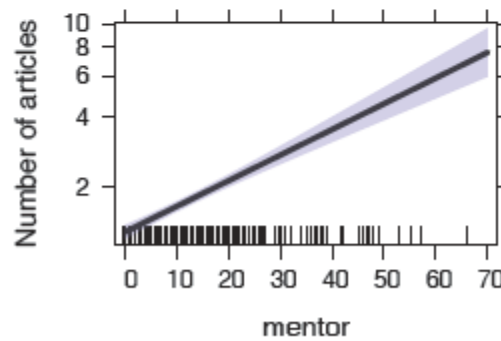
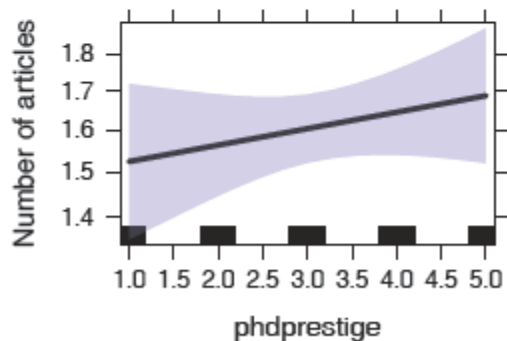
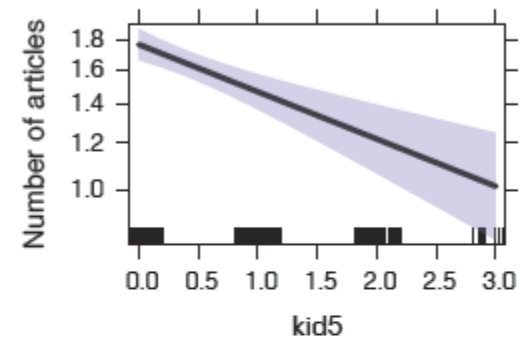
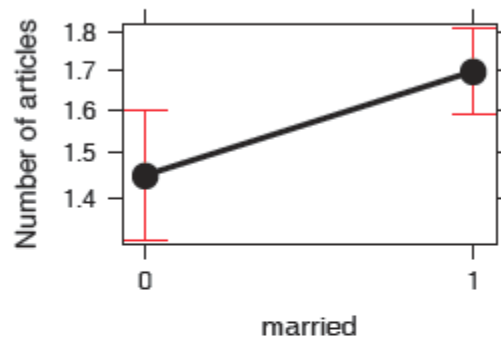
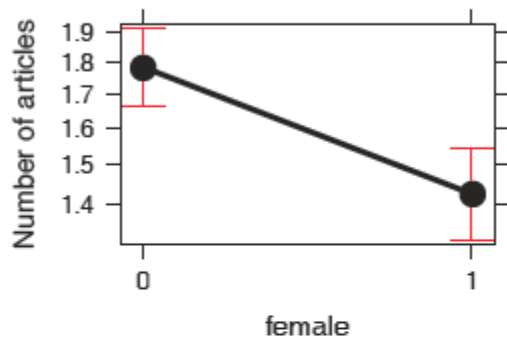
```
# predictors: female, married, kid5, phdprestige, mentor
phd.pois <- glm(articles ~ ., data=PhdPubs, family=poisson)
phd.nbin <- glm.nb(articles ~ ., data=PhdPubs)

LRstats(phd.pois, phd.nbin)

## Likelihood summary table:
##           AIC   BIC LR Chisq  Df Pr(>Chisq)
## phd.pois 3313 3342    1634  909    <2e-16 ***
## phd.nbin 3135 3169    1004  909    0.015 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

# Looking ahead: Effect plots

Effect plots show the predicted values for each term in a model, averaging over all other factors.



These are better visual summaries for a model than a table of coefficients.

# Summary

- Discrete distributions are the building blocks for categorical data analysis
  - Typically consist of basic counts of occurrences, with varying frequencies
  - Most common: binomial, Poisson, negative binomial
  - Others: geometric, log-series
- Fit with `goodfit()`; plot with `rootogram()`
  - Diagnostic plots: `Ord_plot()`, `distplot()`
- Models with predictors
  - Binomial → logistic regression
  - Poisson → poisson regression; logliner models
  - These are special cases of **generalized** linear models